
On the Value of Abstractions: Abstraction Selection in Bounded Program Synthesis

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Abstract

Library learning allows a computational solver to use its experience solving problems to add abstractions to its library that aid in future problem solving. How it chooses those abstractions, however, affects its performance on unseen test problems. Additionally, the value of each abstraction a solver adds to its library is contextually dependent on factors like: its search process, search budgets, and library size. In this paper, we explore two different approaches for selecting abstractions and test the performance of the resulting libraries they construct. Namely, we compare **compression**—which chooses abstractions that compress the size of a set of previously-found past solution programs by minimizing the number of subprograms across the entire set—against **search cost minimization**—which chooses abstractions that minimize solution search cost on a new distinct problem set. Both selection procedures choose abstractions from the same set of subprogram candidates, created by initially finding solution programs using only primitive substates and operations that are permitted by a starter domain specific language (DSL). We used Pattern Builder for this study, which presents a solver with a pattern on a 10x10 pixel grid and tasks it to build a program using its library that constructs that pattern. Asking how a solver should optimally add subprograms to its library is worthwhile not only because it can improve held-out problem-solving performance but also because the procedures by which abstractions are chosen differ in computational cost: any increased performance should justify the computational cost required to achieve it. This paper also explores how library size affects solver performance under both abstraction selection procedures under a fixed search budget—and how a search budget interacts with library size and test performance. Each subprogram added to a library increases the number of possible shallow-depth programs a solver must search through, which can exhaust its search budget. While abstractions themselves reduce search by not requiring solvers to reconstruct their subprograms, it’s possible that this search reduction is only used deeper into a program trajectory. Such cases may exhaust a search budget before a solution program is found, highlighting a tradeoff between library size and search cost.

In this study, we found that with 10 abstractions, both selection procedures outperformed the primitive-only DSL. Search cost minimization outperformed compression by solving 4.08 more test problems out of 100 (95 percent CI 2.46 to 5.71). At 30,000 program attempts, mean performance peaked at 11 abstractions but then fell sharply, ultimately underperforming the primitive-only DSL at 20 added abstractions. Interestingly, both selection procedures’ library performance dropped sharply from library size 1 to 2 abstractions. We determined that this performance drop was caused by the 30,000 program search budget: after holding the libraries constant and increasing allowed search, all of the unsolved problems that constituted the performance reduction were recovered. These findings strongly support the view the value of a library abstraction depends on the context in which it’s used and the constraints a solver is subject to.

1 Introduction

Pattern Builder provides the program-synthesis task in this paper. Each problem presents a solver with a pattern on a binary 10×10 pixel grid, and the solver must build a program that constructs that pattern. It starts with no input grid and combines six primitive grids with fixed operations until a program’s output matches the target. Figure 1 shows one target and one program that builds it. The operation `add` returns the union of two grids.

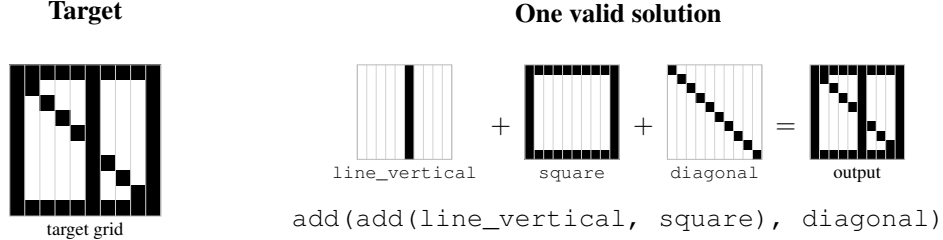


Figure 1: A grid target and one expression that reproduces it.

We call each target-grid task a problem T_i and write its problem state as S_i^{problem} . A solution program P_i transforms that problem state into a solution state S_i^{solution} that matches the target grid:

$$S_i^{\text{problem}} \xrightarrow{P_i} S_i^{\text{solution}}.$$

A solution program is a composite function, possibly nested, and its parts are *subprograms*. The intermediate grids a program produces while it executes are *substates*. Operations transform substates in two ways. A unary operation f_u applies directly to one substate and produces another, while a binary operation f_b combines two substates and produces another:

$$S_1 \xrightarrow{f_u} S_2, \quad (S_1, S_2) \xrightarrow{f_b} S_3.$$

Figure 2 shows how substates, operations, and subprograms compose into a solution program.

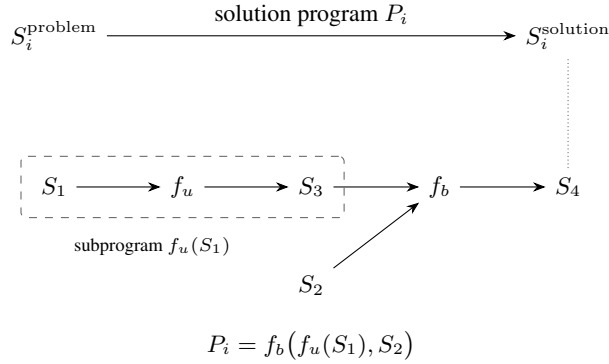


Figure 2: One solution program from the inside. The program P_i maps the problem state to the solution state. Within it, the unary operation f_u transforms substate S_1 into substate S_3 , and the binary operation f_b combines S_3 with S_2 to produce S_4 , which is the solution state. The fragment $f_u(S_1)$ is one subprogram of P_i .

The solver does not know which program will construct the target, so it searches. At each substate, it can apply a unary operation or combine that substate with another through a binary operation, and it works through these possible operations and transitions until a program’s output matches the target.

The solver first solves problems with primitives only. We call those problems the primitive-only problem set, T_{prim} ; it exists to give the solver experience before any abstractions are available. The successful programs the solver finds on it form the primitive-only solution program set,

$$P_{\text{prim}} = \{P_{1,\text{prim}}, \dots, P_{n,\text{prim}}\},$$

where each $P_{i,\text{prim}}$ solves its problem T_i . This set is the solver's empirical experience, and it is the only place we look for abstractions: every subprogram found within a primitive-only solution program is an *abstraction candidate*. Together, the candidates form the set

$$A_{\text{candidates}} = \bigcup_{P \in P_{\text{prim}}} \text{subprograms}(P).$$

Figure 3 traces this path from problems to solution programs to candidates.

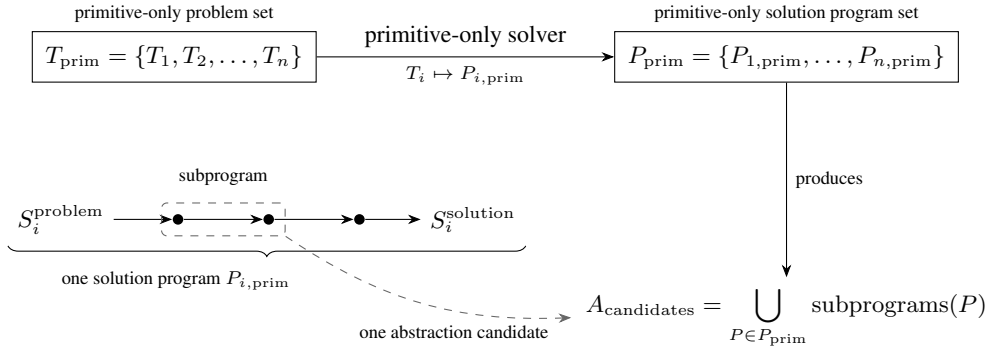


Figure 3: From solver experience to abstraction candidates. The primitive-only solver solves each problem T_i in T_{prim} , and the solution programs it finds form P_{prim} , the solver's experience. Every subprogram found within those programs, such as the marked fragment of $P_{i,\text{prim}}$, enters the abstraction candidate set $A_{\text{candidates}}$.

An *abstraction* is a selected subprogram stored as one item in the solver's library. During search, the solver can use a stored abstraction directly, without reconstructing its internal subprograms. A selector decides which candidates earn that place: it chooses a subset A_q^* of $A_{\text{candidates}}$ and adds it to the primitive library L_{prim} , producing the library

$$L_q = L_{\text{prim}} \cup A_q^*,$$

where q names the selector. This leads to the fundamental question of the paper: how should a solver select abstractions from its experience if its goal is maximum problem-solving performance on an unseen test problem set?

We compare two selectors that answer this question differently. **Compression** chooses the abstractions that minimize the number of subprograms needed to represent the primitive-only solution programs, so it works entirely from solutions the solver has already found. **Search cost minimization** instead measures candidates in use: it chooses the abstractions that empirically minimize search cost on the *Search Cost Minimization Problem Set*, T_{SCM} , a new problem set that exists only to supply that measurement. Selecting from the same candidates, the two selectors produce the selected abstraction sets $A_{\text{compression}}^*$ and $A_{\text{search cost}}^*$, and with them the libraries $L_{\text{compression}}$ and $L_{\text{search cost}}$. To measure which library is better, we reserve the unseen test problem set, T_{test} : a set of problems no selector ever sees, so performance on it shows how well the selected abstractions generalize. The same solver attempts the same test problems with each library. Figure 4 lays out this comparison; the two paths differ only in the selection objective and the data it consults.

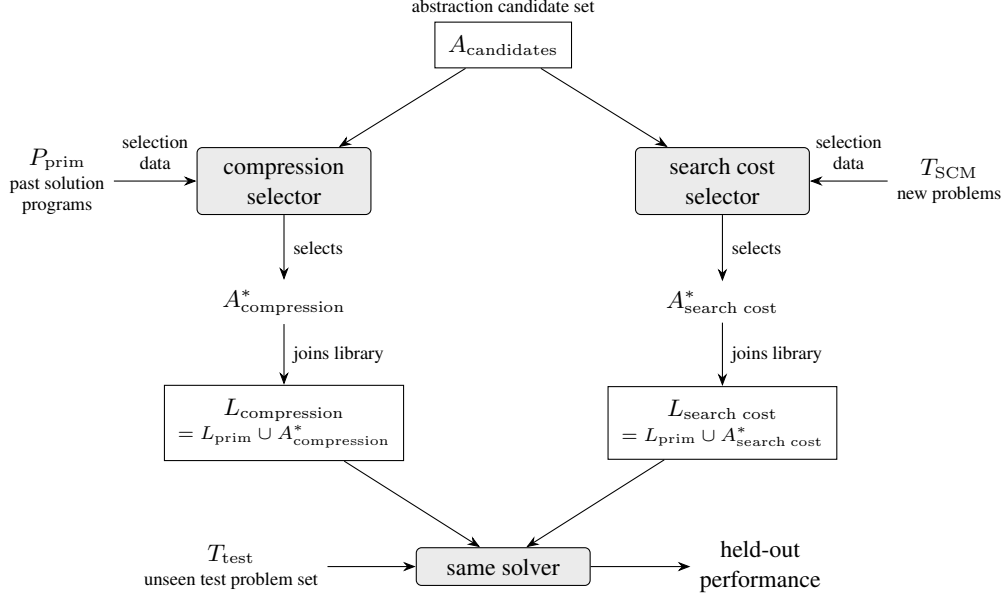


Figure 4: Two selection paths from one candidate set. Both selectors choose from $A_{\text{candidates}}$, but compression scores candidates against the primitive-only solution programs, while search cost minimization scores them by empirical search cost on the Search Cost Minimization Problem Set. Each selected set joins the primitive library, and the same solver then attempts the same unseen test problems with each resulting library.

For example, the union of `line_vertical` and `square` in Figure 1 is one abstraction candidate. Storing it can shorten a later program, yet it also becomes one more item the solver must consider at every substate during search. Under a fixed search budget, a larger library can spend its attempts on shallow programs before the solver ever reaches the deeper program where an abstraction pays off. An abstraction’s value therefore depends on library size, the search process, and the search budget.

Selection itself has a cost, and the two selectors pay different amounts. Compression reuses solution programs the solver already has, while search cost minimization requires the solver to work through the Search Cost Minimization Problem Set before it selects anything. This gap is part of why the comparison is worthwhile: any increase in held-out performance should justify the computational cost required to achieve it.

Experiment I compares the selectors while changing the similarity between the selection problems and the unseen test problems, which reduces shared motifs as a source of performance. Experiment II changes the number of selected abstractions under a fixed search budget. Experiment III holds the libraries fixed and changes the search budget. Together, the three experiments test how selection objective, library size, and available search shape the value of an abstraction.

2 Related Work

Library learning adds reusable program components to a solver’s library so that later synthesis becomes easier, and many systems choose those components by compressing known solution programs. DreamCoder learns a function library together with a neural search policy (Ellis et al., 2021). Stitch finds the functions that minimize the cost of a rewritten program corpus (Bowers et al., 2023). A recent study used Pattern Builder to examine which helper programs people choose as task structure changes (Hernandez Cano et al., 2026); their choices were consistent with compression of expected future solutions. Our study instead measures how automated abstraction selection changes bounded search.

Iba described the search tradeoff that motivates this measurement: macros shorten solution paths and increase the search branching factor (Iba, 1989). AbstractBeam later added Stitch abstractions to guided bottom-up synthesis and improved performance on related tasks, but a distribution shift removed the benefit, and unused abstractions could enlarge the language (Zenkner et al., 2024). AMaze provides the closest technical comparison: it modifies a domain-specific language (DSL) using predicted synthesis time, estimated from the enumeration rank of solver-specific program fragments, and its whole-program compression baseline slowed two of four synthesizers (Ye et al., 2026). Our work measures capped search cost directly, with exact enumeration over a fixed candidate set within each seed. We vary library size, change only the search budget for fixed libraries, and measure the work each selection procedure spends to select its library. These controls test how selection objective, library size, and search budget jointly determine an abstraction’s value.

3 Methodology

3.1 Experiment Overview

We test how abstraction selection affects bounded-search performance, and later experiments change library size and search budget. Experiment I compares four selection procedures: three compression variants that differ in the solution programs they score, and search cost minimization. Each procedure selects from the same abstraction candidate set, and we then test each selected library on the same held-out problems.

The three problem sets of the introduction carry the design. The primitive-only problem set T_{prim} contains 100 problems; it creates the abstraction candidate set and supplies the solution programs that two compression variants score. The Search Cost Minimization Problem Set T_{SCM} contains 25 new problems; search cost minimization scores search cost on it directly, and one compression variant scores solution programs acquired for it. The unseen test problem set T_{test} contains 100 problems that no procedure sees; we reserve it for held-out evaluation. We repeat the experiment across 30 random problem worlds. Each seed creates one world with 12 hidden motifs, the grids from which the generator builds every target, and one primitive-only problem set shared by six primary conditions, each with a separate T_{SCM} and T_{test} . Table 1 gives the size and purpose of each problem set.

Table 1: Problem sets and their use.

Problem Set	Size	Purpose
Primitive-only problem set (T_{prim})	100	Creates the abstraction candidate set and supplies solution programs for compression.
Search Cost Minimization Problem Set (T_{SCM})	25	Supplies the selection data for search cost minimization and one compression variant.
Unseen test problem set (T_{test})	100	Measures each selected library on held-out problems.

Each seed has one abstraction candidate set of 50 candidates, and every selection procedure selects from it. Table 2 defines the four selection procedures and the labels that later equations use. The three compression variants score different solution program sets: all of P_{prim} , the subset $P_{\text{prim},25}$ acquired for a fixed 25-problem subset $T_{\text{prim},25} \subset T_{\text{prim}}$, or the set P_{SCM} of solution programs the solver acquires for the Search Cost Minimization Problem Set:

$$T_{\text{SCM}} \xrightarrow{\text{solver}} P_{\text{SCM}}.$$

The fourth procedure, search cost minimization, scores search cost on T_{SCM} directly.

Each compression variant runs a separate target-directed search for each scoring problem, keeps the first solution found, and omits unsolved problems. Compression can only score problems that produced a solution, so compression on P_{SCM} uses a larger 90,000-attempt acquisition budget on

Table 2: Four selection procedures in Experiment I.

Label	Selection procedure	Selection data	Search during selection	Score
comp-100	Compression on all primitive-only solution programs	P_{prim} (100 problems)	Acquire solutions; 30,000 attempts per problem	Remaining operations in acquired solutions
comp-25	Compression on a 25-problem primitive-only subset	$P_{\text{prim},25}$ (25 problems)	Acquire solutions; 30,000 attempts per problem	Remaining operations in acquired solutions
comp-SCM	Compression on solution programs acquired for T_{SCM}	P_{SCM} (25 problems)	Acquire solutions; 90,000 attempts per problem	Remaining operations in acquired solutions
scm	Search cost minimization on T_{SCM}	T_{SCM} (25 problems)	Score each trial library; 30,000 attempts	Mean capped search cost for all 25 problems

T_{SCM} . Search cost minimization runs one size-ordered search for each trial library, whose output record scores all 25 problems of T_{SCM} , and an unreached target receives the maximum cost of 30,000. Hidden generator programs are never selector inputs. A fixed seeded shuffle selects the 25 problems of $T_{\text{prim},25}$, and all six conditions within a seed use the same subset. Across the 180 primary Experiment I cells, acquisition produced solutions for a mean of 62.70 of the 100 problems in T_{prim} , 16.33 of the 25 in $T_{\text{prim},25}$, and 16.59 of the 25 in T_{SCM} .

In Experiment I, each procedure selects ten abstractions. We evaluate every selected library on the same 100 test problems with a 30,000-attempt limit. The primary comparison between search cost minimization and compression on P_{SCM} uses the same 25 problems of T_{SCM} . Search cost minimization scores all 25 outcomes, whereas compression on P_{SCM} scores only acquired solution programs and omits a mean of 8.41 unsolved problems. The comparison holds the candidate set, test problems, library size, and test budget constant, so it compares complete procedures that differ in what they score and whether they include unsolved problems.

3.2 Synthesis and Bounded Search

Each problem T_i presents one binary target grid, the grid its solution state S_i^{solution} must match. The primitive library L_{prim} contains six grids: blank, horizontal line, vertical line, diagonal, square, and triangle. Let L denote the current ordered library. A program is a library item or an operation applied to one or two programs. The unary operations form the set F_u , with $f_u \in F_u$: invert, a left-to-right flip, a top-to-bottom flip, and diagonal reflection. The diagonal grid runs from the upper-left corner to the lower-right corner. Invert changes each black cell to white and each white cell to black. The binary operations form the set F_b , with $f_b \in F_b$: add (union), subtract (set difference), and `overlap` (intersection). Program P produces grid $\text{eval}(P)$ and solves T_i when $\text{eval}(P)$ equals the target grid of T_i . Program size is the number of operation applications. A primitive or stored abstraction has size zero. Let B be the maximum number of candidate programs that the solver can try. The deterministic bottom-up solver tries programs in order of increasing size. It stops after B attempts or after it has tried all programs with seven or fewer operation applications. The solver keeps the first program for each distinct output grid. The implementation fixes the order of operations, child-program sizes, and commutative arguments. Each generated program counts as one search attempt. If it produces an output grid that the solver has already seen, the solver discards it after counting the attempt. For search cost minimization scoring and final evaluation, one size-ordered search records all outputs for a library and the first attempt that produced each target. Compression uses separate target-directed searches to acquire solution programs.

Search cost is a function of a problem, a library, and a budget:

$$(T_i, L, B) \xrightarrow{\text{solver}} N_B(T_i, L).$$

N_B measures how many attempts the solver needs to reach the target grid of T_i , capped at the budget:

$$N_B(T_i, L) = \begin{cases} n, & \text{if the target grid of } T_i \text{ first appears at attempt } n \leq B, \\ B, & \text{if the solver does not reach it within the budget.} \end{cases} \quad (1)$$

Unless stated otherwise, $B = 30,000$. We use the number of held-out test targets reached among 100 as our primary outcome.

3.3 The Abstraction Candidate Set

The introduction builds the abstraction candidate set through the chain

$$T_{\text{prim}} \xrightarrow{\text{primitive-only solver}} P_{\text{prim}} \xrightarrow{\text{subprograms}(\cdot)} A_{\text{candidates}}.$$

The full set of subprograms is too large to score repeatedly, so the experiment bounds $A_{\text{candidates}}$ to 50 candidates in two stages. Each candidate is a subprogram. If selected, the solver stores the candidate’s output grid as a size-zero library item and can then use that grid without rebuilding the subprogram from primitives. The candidate set therefore defines the available library modifications, while the selection procedures determine which modifications are made.

The first stage collects reached grids. The primitive-only solver uses the standard 30,000-attempt budget and records the first program found for each distinct output grid. We exclude the blank grid and the six primitive grids because the library already contains them. We keep grids whose first programs have one to four operations, which yields nontrivial candidates while limiting the initial pool to small subprograms.

The second stage keeps candidates with observed potential for reuse. We apply the same one-step test to every target grid of T_{prim} separately. A candidate supports a target if one additional invert or reflection can produce that target from the candidate’s grid, or if one add or subtract involving another reached grid can do so. The support count is the number of distinct targets that satisfy this test. We retain only candidates with a support count of at least two, the smallest count that indicates potential reuse across problems.

We rank the retained candidates by decreasing support count, then break ties by the attempt at which the primitive-only solver first reached the grid and by program string. The top 50 candidates form the experiment’s $A_{\text{candidates}}$. This cap keeps repeated candidate evaluation tractable, while the fixed ranking and tie rules make construction deterministic. Construction uses no problems from T_{SCM} or T_{test} , and every selection procedure within a problem world receives the same fixed candidate set. The next subsection describes how each procedure selects abstractions from $A_{\text{candidates}}$, one at a time.

3.4 Selection Procedures

Each selection procedure builds its library one abstraction at a time. A selector takes the candidate set, its own selection data, and a target library size K , and returns the selected abstraction set:

$$\begin{aligned} \text{Select}_{\text{compression}}(A_{\text{candidates}}, P_{\text{score}}, K) &\longrightarrow A_K^*, \\ \text{Select}_{\text{search cost}}(A_{\text{candidates}}, T_{\text{SCM}}, B, K) &\longrightarrow A_K^*. \end{aligned}$$

Here P_{score} is the solution program set that a compression variant scores: P_{prim} , $P_{\text{prim},25}$, or P_{SCM} , each containing the first solution found for each solved scoring problem.

Write $A_k^* = \{a_1, \dots, a_k\}$ for the selected abstraction set after k selections, with $A_0^* = \emptyset$; the subscripts on $a_1, \dots, a_k$ record the order of selection. Order matters during search, so the ordered library $L(A_k^*)$ lists the six primitive grids and then the stored grid of each abstraction, in selection

order, and a candidate under consideration takes the last position. The resulting library of procedure q after K selections is $L_q^{(K)} = L(A_K^*)$, which as a set of items is the introduction's $L_q = L_{\text{prim}} \cup A_q^*$.

Selection is greedy. Each step uses an objective J to score the selected set extended by each unselected candidate a , keeps the candidate with the lowest cost, and never revisits earlier selections:

$$\begin{aligned} a_k &= \arg \min_{a \in A_{\text{candidates}} \setminus A_{k-1}^*} J(A_{k-1}^* \cup \{a\}), \\ A_k^* &= A_{k-1}^* \cup \{a_k\}, \quad k = 1, \dots, K. \end{aligned} \quad (2)$$

The two objectives differ in what they score. Compression scores the acquired solution programs in P_{score} . For program P , let $\text{children}(P)$ be its ordered list of immediate child programs. A subtree whose output grid matches a stored abstraction becomes one size-zero library item, so its internal operations count zero. For a selected set A^* , write $\text{eval}(A^*)$ for its set of stored grids, $\{\text{eval}(a) : a \in A^*\}$. The number of remaining operations is

$$n_{\text{ops}}(P, A^*) = \begin{cases} 0, & \text{if } \text{children}(P) = () \text{ or } \text{eval}(P) \in \text{eval}(A^*), \\ 1 + \sum_{P' \in \text{children}(P)} n_{\text{ops}}(P', A^*), & \text{otherwise.} \end{cases} \quad (3)$$

Each operation application is one composite subprogram, so counting remaining operations counts the subprograms still needed to represent the scored solutions. Search cost minimization instead scores each trial library in use, through the capped search cost on T_{SCM} . The two objectives are

$$\begin{aligned} J_{\text{compression}}(A^*) &= \sum_{P \in P_{\text{score}}} n_{\text{ops}}(P, A^*), \\ J_{\text{search cost}}(A^*) &= \frac{1}{|T_{\text{SCM}}|} \sum_{T_i \in T_{\text{SCM}}} N_B(T_i, L(A^*)). \end{aligned} \quad (4)$$

$J_{\text{compression}}$ counts the operations that remain across the scored solutions after stored grids replace their subtrees. $J_{\text{search cost}}$ averages the capped search cost over T_{SCM} , and a problem the solver does not reach within B attempts costs the full budget.

Tie rules keep both selectors deterministic. If two candidates have the same compression cost, compression compares their text forms in lexicographic order, character by character from left to right. For example, `add(diagonal, square)` comes before `add(line_vertical, square)` because `d` comes before `l`. If two trial libraries have the same search cost, search cost minimization first selects the library that solves more problems of T_{SCM} and applies the same text-order rule if both solve the same number.

3.5 Problem Sets and Problem-Set Similarity

The problems a procedure works through during selection can act as additional training for the solver, so part of any advantage on the unseen test problems could come from exposure to similar problems. We vary problem-set similarity to control for this: when the test problems share little structure with the primitive-only problems that supply the candidates and the compression solutions, exposure helps less, and the comparison comes closer to measuring the selection objective alone.

Every target grid is composed of substates, which we call motifs, and each motif is a grid that a subprogram of two, three, or four operations can reach. Each seed defines 12 hidden motifs, four at each of those sizes. The generator builds every problem from them: it samples two or three distinct motifs, combines them from left to right with uniformly sampled binary operations and up to two unary operations, and rejects invalid and duplicate grids, with fixed schedules setting the motif and unary-operation counts. We can therefore set how often each motif appears in a problem set through its sampling probability, and two problem sets are similar when the same motifs are common in both. Selection procedures receive only target grids or acquired solutions, never motif identities or hidden programs.

T_{prim} uses motif probabilities $\mathbf{p}_{\text{prim}}$. To build a dissimilar set, we keep the same probability values and reassign them to different motifs, which makes different motifs common while leaving the mix of common and rare unchanged. Reversal is the farthest reassignment: it changes $(p_1, \dots, p_{12})$ to $(p_{12}, \dots, p_1)$, so the most common motifs become the most rare. Reversal is still one fixed rearrangement, though, and an effect that appears under it could depend on that exact opposite ordering, so a random permutation supplies a second alternative: it assigns the same values to motifs at random, dissimilar without being systematically opposite. A shift parameter $\alpha \in \{0, 0.5, 1\}$ then blends the two distributions,

$$\mathbf{p}(\alpha) = (1 - \alpha) \mathbf{p}_{\text{prim}} + \alpha \mathbf{p}_{\text{alt}},$$

so similarity varies in steps: the same frequencies at $\alpha = 0$, an equal mixture at $\alpha = 0.5$, and the reassigned frequencies at $\alpha = 1$. In each primary condition, T_{SCM} and T_{test} are separate draws from $\mathbf{p}(\alpha)$. Combining two alternatives with three values of α gives six conditions per seed, and the two conditions at $\alpha = 0$ use separate target draws.

The assigned α sets the intended similarity, and each random draw realizes it imperfectly, so we also measure realized problem-set similarity, ρ , between T_{prim} and T_{test} as the Spearman correlation between their motif-count vectors. We use ρ only for analysis because it requires hidden motif identities and the completed test set.

3.6 Experiments

All three experiments measure the same outcome. Let q name a selection procedure, using the labels of Table 2, and let $i \in \{1, \dots, 30\}$ name a seed and $c \in \{1, \dots, 6\}$ a problem-set similarity condition. The outcome of one cell is the number of test problems that the resulting library solves with its first K abstractions and budget B :

$$(L_q^{(K)}, T_{\text{test}}, B) \xrightarrow{\text{solver}} Y_{ic}^q(K, B) \in \{0, \dots, 100\}.$$

A seed’s six conditions share one world, so we average them into one seed score and then average the 30 seed scores:

$$\bar{Y}_i^q(K, B) = \frac{1}{6} \sum_{c=1}^6 Y_{ic}^q(K, B), \quad \hat{\mu}^q(K, B) = \frac{1}{30} \sum_{i=1}^{30} \bar{Y}_i^q(K, B). \quad (5)$$

The experiment mean $\hat{\mu}^q(K, B)$ therefore equals the mean of all 180 cells. The conditions have equal weight, but only the 30 seeds are independent.

Experiment I: Which Abstractions? Experiment I asks which selection objective produces the better library when both selectors receive the same candidates and the same selection problems, and whether problem-set similarity changes the difference between search cost minimization and compression on P_{prim} . Thirty seeds under six primary conditions gave 180 cells, and each cell used $K = 10$, one candidate set, and the same held-out test targets for all procedures. Compression on P_{prim} serves as a practical comparator because it scores every solution the solver already has. A further 30 registered stale-SCM cells drew T_{SCM} at $\alpha = 0.5$ and test problems at $\alpha = 1$; they stress test outdated selection data, and we excluded them from the primary estimates because those estimates require T_{SCM} and test problems from the same distribution. Using the experiment mean $\hat{\mu}^q(K, B)$ defined above, we registered two contrasts at $K = 10$ and $B = 30,000$:

$$\begin{aligned} & \hat{\mu}^{\text{scm}}(10, 30,000) - \hat{\mu}^{\text{comp-SCM}}(10, 30,000), \\ & \hat{\mu}^{\text{comp-SCM}}(10, 30,000) - \hat{\mu}^{\text{comp-25}}(10, 30,000). \end{aligned} \quad (6)$$

A positive value favors the first procedure in each difference. The first contrast compares the two complete procedures that share T_{SCM} . The second compares two complete compression procedures that differ in scoring-problem source and acquisition budget. We hypothesized that compression

on primitive-only solutions would improve relative to search cost minimization as the motif frequencies of T_{prim} and T_{test} became more similar, because repeated motifs can let abstractions from T_{prim} help test problems. The test uses both similarity measures from the previous subsection: for the assigned shift, we averaged the two alternative assignments and calculated each procedure difference at $\alpha = 1$ minus $\alpha = 0$; for realized similarity, regressions related these differences to ρ with seed fixed effects and seed-clustered standard errors. We also measure what each selection costs, so its performance can be weighed against the work spent to achieve it. Let W_q count, within one seed-condition cell, the candidate-program attempts used to select procedure q 's library. Search cost minimization scores 455 = 50 + 49 + $\dots$ + 41 trial libraries with at most 30,000 attempts each. The work $W_{\text{comp-SCM}}$ includes acquiring solutions for T_{SCM} , while $W_{\text{comp-100}}$ treats the prior acquisition of P_{prim} as a sunk cost; both counts exclude compression segmentation, the recomputation of remaining operations that scores each trial. Let $\bar{N}_q$ be the mean capped cost across that cell's 100 held-out test problems. For comparator $q \in \{\text{comp-SCM}, \text{comp-100}\}$, $n_{\text{pay}}(q)$ is the number of future problems needed to recover search cost minimization's additional selection work:

$$n_{\text{pay}}(q) = \begin{cases} \left\lceil \frac{\max(0, W_{\text{scm}} - W_q)}{\bar{N}_q - \bar{N}_{\text{scm}}} \right\rceil, & \text{if } \bar{N}_q > \bar{N}_{\text{scm}}, \\ \infty, & \text{if } \bar{N}_q \leq \bar{N}_{\text{scm}}. \end{cases} \quad (7)$$

This post hoc calculation assumes constant mean savings and uses only program attempts. Separately, search cost minimization also scored the 25 problems of $T_{\text{prim},25}$ post hoc, which lets the indirect-compression analysis in the results compare the two objectives on shared scoring problems.

Experiment II: Library Size. Experiment II asks whether adding more abstractions keeps helping under a fixed search budget. Thirty fresh seeds (6541–6570) and six conditions formed 180 cells. Compression on P_{prim} , compression on P_{SCM} , and search cost minimization each selected 20 abstractions in one selection order, and we evaluated every prefix A_K^* from $K = 0$ to $K = 20$ with $B = 30,000$, so one selection run per procedure covers the whole size range. The value $K = 0$ is the primitive library. For procedures q and q' at $B = 30,000$, their performance difference at library size K is

$$D_{q,q'}(K) = \hat{\mu}^q(K, B) - \hat{\mu}^{q'}(K, B), \quad (8)$$

the number of additional test problems that procedure q solves at each library size. We registered three method contrasts: $D_{\text{scm}, \text{comp-SCM}}(20)$, $D_{\text{scm}, \text{comp-100}}(20)$, and the change in $D_{\text{scm}, \text{comp-SCM}}$ from $K = 10$ to $K = 20$. Two additional registered within-procedure contrasts measured the changes in $\hat{\mu}^{\text{scm}}$ and $\hat{\mu}^{\text{comp-100}}$ from $K = 10$ to $K = 20$. Observed peaks and the one-to-two drop were descriptive.

Experiment III: Search Budget. Experiment III asks whether the drop from one abstraction to two came from the abstractions themselves or from the search budget. To isolate the budget's effect, Experiment III reused Experiment II's exact $K = 1$ and $K = 2$ libraries and varied only $B \in \{30,000, 45,000, 60,000, 90,000\}$; worlds, targets, abstraction order, size limit, and enumeration order remained fixed, and selection did not run again. Here, q names the fixed library selected in Experiment II. For $q \in \{\text{comp-100}, \text{scm}\}$, let $\Delta_q(B)$ be the effect of adding the second abstraction and I_q its interaction with budget:

$$\begin{aligned} \Delta_q(B) &= \hat{\mu}^q(2, B) - \hat{\mu}^q(1, B), \\ I_q &= \Delta_q(90,000) - \Delta_q(30,000). \end{aligned} \quad (9)$$

A positive I_q means that increasing the search budget raises the $K = 2$ minus $K = 1$ performance difference. We calibrated the budgets on ten earlier seeds (6511–6520), separate from the Experiment II seeds, before we inspected above-30,000 outcomes, so the tested budgets did not depend on the outcomes they would judge.

3.7 Statistical Analysis

Primary comparisons average six related differences between procedures into one seed-level difference. The 30 seed-level differences test whether the advantage repeats across worlds. Similarity analyses keep conditions separate because similarity varies by condition but treat conditions that share a seed as related. The two primary comparisons use two-sided 95% confidence intervals based on Student’s t distribution. Experiment II tracks method differences across $K = 1, \dots, 20$, while Experiment III tests whether more search changes the one-to-two abstraction loss. Both use maximum- t intervals from 10,000 seed-level bootstrap resamples for related estimates. Experiment I registered its contrasts and directions before data collection, but selected the Student’s t interval afterward. We specified the Experiment II analysis before collecting fresh data; its results motivated both a registered Experiment III hypothesis that a larger budget would reduce the one-to-two loss and an increase in the tested budgets. Later analyses are descriptive or post hoc.

4 Results

4.1 Experiment I: Which Abstractions?

With ten abstractions, search cost minimization solved 66.08 test problems and compression on P_{SCM} solved 62.00 (Figure 5); the difference was 4.08 (95% CI [2.46, 5.71]). Both used one candidate set and one T_{SCM} , but they scored different things: search cost minimization scored all 25 outcomes, including failures, whereas compression on P_{SCM} scored only the solutions it acquired, a mean of 16.59 of the 25 problems, and omitted the remaining 33.6%. The remaining comparisons place this result among the other libraries. The primitive library solved 58.91 problems, and compression on P_{prim} , which used all 100 primitive-only problems, solved 64.53. Compression on P_{SCM} solved 0.26 fewer problems than compression on $P_{prim,25}$ (95% CI [-1.46, 0.95]). Search cost minimization’s 1.55-problem advantage over compression on P_{prim} remained uncertain because its 95% CI [-0.27, 3.37] included zero. The separate stale-SCM stress test asks whether outdated selection data erases these differences. The means kept the same order: 65.60 for search cost minimization, 61.00 for compression on P_{SCM} , and 65.40 for compression on P_{prim} .

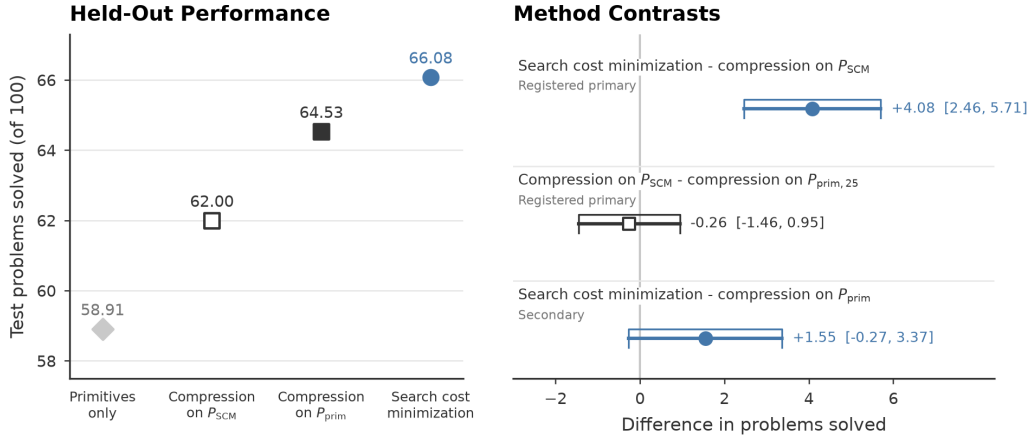


Figure 5: Experiment I held-out results with ten abstractions. The left graph shows means, while the right graph shows paired differences with 95% Student’s t intervals for 30 seeds. Both procedures use the same problems of T_{SCM} . Search cost minimization includes failures; compression on P_{SCM} omits unsolved targets. The second contrast compares compression on P_{SCM} with compression on $P_{prim,25}$.

Selection itself costs search, and later search savings must recover that cost for a more expensive selection method to be useful. Search cost minimization has a higher initial search cost than compression because it performs one bounded search for each trial library and scores all 25 problems of T_{SCM} from its output record. We therefore estimate a break-even point: the number of future problems whose search savings recover that higher initial cost. The median break-even point was several thousand future problems, and some seed-condition cases never reached it. Figure 6 gives the exact selection costs, median break-even points, and case counts.

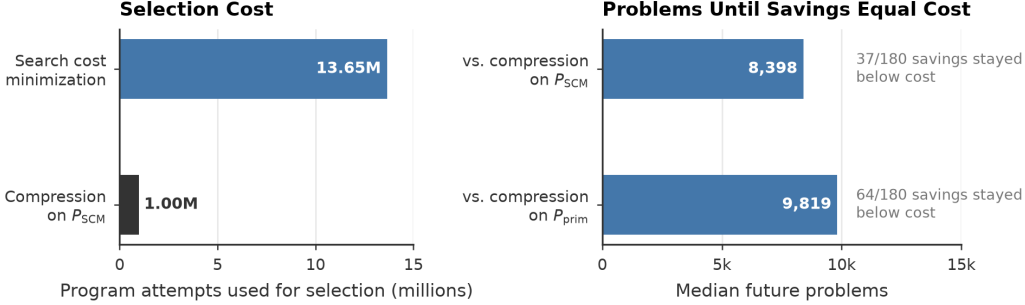


Figure 6: Selection cost and break-even estimates. Gray labels mark cases with no break-even point. Counts are program attempts only; they omit the acquisition of P_{prim} , wall-clock time, and segmentation.

We expected compression to benefit from similar motifs in T_{prim} and T_{test} , and the means followed that direction: search cost minimization minus compression on P_{prim} decreased as problem-set similarity increased, from 2.7 problems for different sets to 1.5 for moderately similar sets and 0.5 for similar sets (left graph in Figure 7). The different-set interval excluded zero, showing search cost minimization’s advantage in that group. The formal tests follow the two similarity measures. By assigned shift, moving from $\alpha = 0$ to $\alpha = 1$ changed search cost minimization minus compression on P_{prim} by 2.10 problems (95% CI $[-0.12, 4.32]$) and search cost minimization minus compression on P_{SCM} by 1.70 (95% CI $[-0.56, 3.96]$). By realized similarity, the seed-controlled slope against ρ was -1.44 for the P_{prim} difference (95% CI $[-3.18, 0.30]$) and -0.36 for the P_{SCM} difference (95% CI $[-2.43, 1.71]$). These intervals all included zero, so the formal tests did not establish the hypothesized effect. The two objectives could also overlap: abstractions chosen for search cost might compress solutions as a side effect. We tested this by applying each library post hoc to one fixed solution set, the solutions acquired for the 25-problem primitive-only subset (right graph in Figure 7). Direct compression on those problems removed 25.78% of operations; search cost minimization scoring the same 25 primitive-only problems removed 12.32%; search cost minimization scoring the 25 problems of T_{SCM} removed 9.79%. Search cost minimization shortened solutions indirectly, but direct compression removed more.

4.2 Experiment II: Library Size

Held-out performance did not change steadily as library size increased (Figure 8). From ten to twenty abstractions, compression on P_{prim} lost 12.35 problems (simultaneous 95% CI $[-13.89, -10.81]$), and search cost minimization lost 12.11 (simultaneous 95% CI $[-13.20, -11.02]$). At twenty abstractions, both libraries solved fewer problems than the primitive library. Between procedures, search cost minimization exceeded compression on P_{SCM} by 5.13 problems (simultaneous 95% CI $[3.45, 6.80]$) and compression on P_{prim} by 1.73 (simultaneous 95% CI $[-0.17, 3.63]$), and its advantage over compression on P_{SCM} increased by 1.85 from ten to twenty (simultaneous 95% CI $[-0.03, 3.73]$). Thus, library size had a larger effect on absolute performance than on method differences.

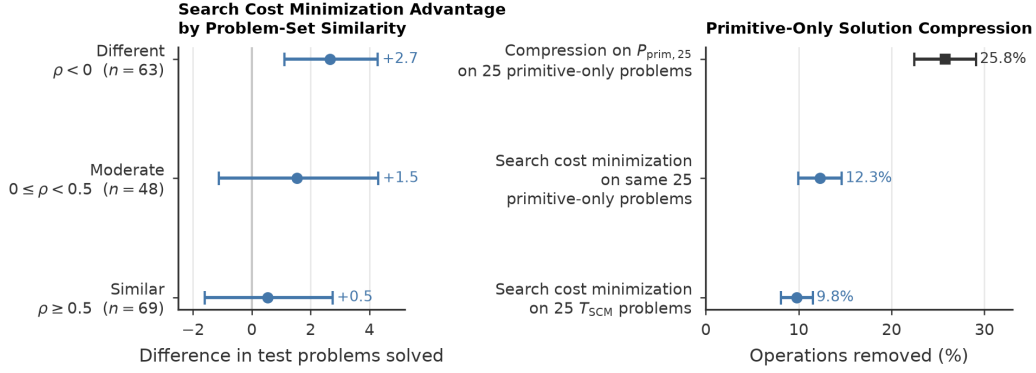


Figure 7: Problem-set similarity and indirect compression. Left: search cost minimization minus compression on P_{prim} at three problem-set similarity levels. Right: post hoc operations removed from the same primitive-only solutions. Intervals are 95% seed-cluster and seed bootstrap, respectively; problem-set similarity bands were not registered.

At $B = 30,000$, the descriptive means were highest at $K = 11$: 66.32 for compression on P_{prim} and 67.85 for search cost minimization. From $K = 1$ to $K = 2$, compression on P_{prim} lost 8.08 problems, and search cost minimization lost 6.14, a drop that Experiment III investigates. For compression on P_{prim} , operations removed from primitive-only solutions increased from 24.14% at $K = 11$ to 27.21% at $K = 20$, while held-out performance decreased over the same range, so more compression did not ensure better bounded search.

4.3 Experiment III: Search Budget

An increased budget reversed the one-to-two decrease for the fixed libraries (Figure 8). As the budget increased from 30,000 to 90,000 program attempts, the second-abstraction effect increased by 11.42 solved problems for compression on P_{prim} (simultaneous 95% CI [10.17, 12.66]) and by 9.61 problems for search cost minimization (simultaneous 95% CI [8.52, 10.69]). At $B = 90,000$, the effects were 3.33 for compression on P_{prim} and 3.47 for search cost minimization. All seed-level interactions were positive. The share of enumerations that reached size-four programs shows whether the budget let the solver reach the larger programs where an abstraction can help: with two abstractions, it increased from 8.1% at 30,000 attempts to 99.2% at 45,000 and 100% at 60,000. By $B = 90,000$, the solver recovered all test instances that the two-abstraction library lost relative to the one-abstraction library at $B = 30,000$. Only the budget changed, so the 30,000-attempt budget caused the one-to-two decrease for these fixed libraries. This result does not set a sufficient budget for other libraries, solvers, or benchmarks.

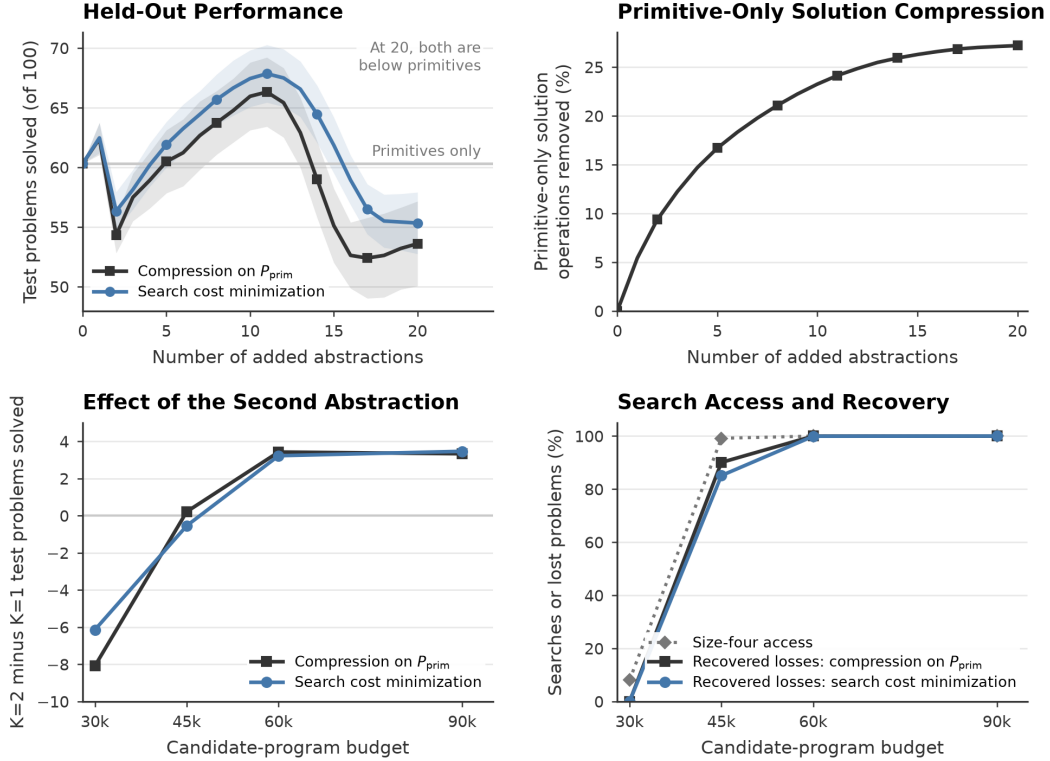


Figure 8: Experiments II and III. The top row shows held-out performance and the compression of primitive-only solutions across library sizes. The held-out bands are simultaneous 95% intervals. The compression graph is post hoc. At $K = 0$, the library has only primitives. The bottom row shows the fixed-library budget test, the share of enumerations that reached size-four programs, and the descriptive recovery of lost problems.

5 Discussion

This paper’s main conclusion is that an abstraction’s value depends on the context in which the solver uses it: the problems and scoring rule used for selection, the library’s contents, future problems, the solver, the search budget, the selection cost, and expected reuse. The conclusion matters because many systems select abstractions by compressing known solution programs, and prior work described the tradeoff a compression score can hide: stored items shorten solutions while enlarging the search (Iba, 1989). Our experiments measure that tradeoff directly, and the sharpest case is Experiment II, where more compression came with worse bounded search.

Experiment I compares the complete selection procedures. Search cost minimization minimized capped search cost for all 25 problems of T_{SCM} , including failures, whereas compression on P_{SCM} minimized operations in acquired solutions for the same problems but omitted unsolved ones. With ten abstractions, search cost minimization solved 4.08 more held-out test problems than compression on P_{SCM} , which supports the hypothesis that it selects more useful abstractions when both procedures use the same selection problems. Scoring libraries in use also charges for the programs each abstraction adds, so search cost minimization can favor lower net search cost. However, the 4.08-problem difference depends in part on which problems of T_{SCM} produced solutions, so it does not isolate scoring-rule effects. Search cost minimization’s 1.55-problem advantage over compression on P_{prim} remained uncertain; compression on P_{prim} scored four times as many problems, and

solutions from more problems could explain the smaller difference, which this experiment did not test.

The match between selection and future problems could change abstraction value, so we hypothesized that compression on primitive-only solutions would become more competitive as the motif frequencies of T_{prim} and T_{test} became more similar. The means followed this pattern: search cost minimization’s advantage over compression on P_{prim} fell from 2.7 problems for different sets to 0.5 for similar sets. However, the formal intervals included zero, so the formal comparisons remain inconclusive. This direction is plausible because compression rewards stored grids that replace sub-trees in primitive-only solutions, and recurring motifs can let those grids shorten test solutions. Our problem-set similarity measure, though, ranks only motif counts and omits operation context, exact grid reuse, program position, and search order. Similar motif counts do not ensure similar search value.

Experiment II showed that abstraction value depends on library size. At 30,000 attempts, observed mean performance was highest at 11 abstractions and fell below the primitive library at 20. Compression on P_{prim} removed more operations from primitive-only solutions as its library grew from 11 to 20 abstractions, even as held-out performance decreased. Each abstraction is a size-zero item whose combinations with operations and other library items create new low-operation programs. The solver examines them first, so they can use the budget before it reaches a larger program where the abstraction helps. A new item also changes when the solver reaches programs that use earlier items. Thus, operation removal measures a shortcut within known solutions but does not measure the search required to reach those solutions.

Search budget can determine whether this added search prevents a solution or only delays it. Experiment III changed only the budget for fixed one- and two-abstraction libraries, and the larger budget reversed the one-to-two loss for both procedures, supporting the hypothesis that a larger budget reverses the loss for those libraries. The size-four diagnostic gives descriptive support for the search-cutoff explanation.

Selection cost creates a separate tradeoff because future savings must recover the initial cost. Search cost minimization required a bounded search for every trial library, and among cases with a break-even point, the median required thousands of future problems. Some experimental cases never reached that point. Compression starts cheaper when representative solutions already exist.

6 Limitations and Future Work

Experimental controls make changes in bounded search easier to identify, but they limit where the results can apply. This study tests one synthetic, target-only grid benchmark. Every abstraction is a stored output grid with size zero, while the primitives, operations, program-size limit, and target generator stay fixed. Learned functions and parameterized abstractions can apply across inputs and produce different search costs. Future work should repeat these experiments on input–output synthesis tasks and in other domains. A direct comparison of stored grids and learned functions should hold the solver and budget fixed.

The search-cutoff explanation also depends on the solver. This solver performs deterministic bottom-up enumeration and orders programs by operation count, so library order and duplicate removal affect when it reaches a target. A guided solver could prune low-value combinations or reach useful larger programs sooner. Future studies should compare deterministic enumeration with guided search and test other costs for abstraction use. A joint sweep can test whether library size and test performance have the same relation under other budgets.

Candidate discovery also limits how broadly the results apply. All procedures select from the same 50 candidates derived from T_{prim} , so search cost minimization cannot select an abstraction outside this set. The candidate set favors grids that support at least two targets of T_{prim} and first appear in programs with one to four operations. Experiment II evaluates prefixes of one greedy sequence rather than optimizing each library size. Future work should vary candidate-set construction and

compare greedy selection with joint selection of library items and their order. A selection objective could also charge for the new programs that each candidate adds.

The primary Experiment I comparison evaluates complete procedures, so it does not isolate scoring-rule effects. Search cost minimization scored all 25 problems of T_{SCM} and their failures, whereas compression on P_{SCM} scored only acquired solutions, so the 4.08-problem gap includes differences in what each procedure scores and whether it includes unsolved problems. Compression on P_{prim} used solutions acquired from 100 primitive-only problems and therefore used more scoring problems from a different set. To isolate scoring-rule effects, a comparison should use fixed target-solution pairs: search cost minimization would score the targets, compression would score their solutions, and solution acquisition would be fixed before data collection. The similarity measure has a separate limit: it requires hidden motif counts from completed test data, so it cannot guide deployment. Future work should define the measure before data collection, vary compositional structure directly, and use more independent seeds to improve precision.

Several findings need independent confirmation. The observed performance peak near 11 abstractions and the one-to-two decrease were descriptive. Experiment III reused the Experiment II seeds and fixed one- and two-abstraction libraries, so its budget effect applies only to those libraries and is not an independent replication. An independent test should use new seeds and libraries. A broader sweep can identify the budgets needed to reach programs that use added abstractions.

Finally, the break-even analysis counts only program attempts, omitting wall-clock time, memory use, and compression segmentation. The comparison with compression on P_{prim} treats the prior acquisition of those solutions as sunk. The calculation assumes constant mean savings across future problems. Future work should measure end-to-end acquisition, selection, and evaluation costs for expected workloads under distribution change.

7 Conclusion

The introduction asked how a solver should select abstractions from its experience if its goal is maximum problem-solving performance on an unseen test problem set. With ten abstractions, search cost minimization outperformed compression on P_{SCM} by solving 4.08 more test problems out of 100, in a comparison of complete procedures. Its advantage over compression on P_{prim} remained uncertain, and mean performance differences across problem-set similarity levels followed the problem-set similarity hypothesis while their formal intervals included zero.

The deeper lesson is that the same abstractions can help or hurt depending on the library around them and the search the solver is allowed. At 30,000 attempts, both procedures improved on primitive-only search with ten abstractions, and mean performance peaked near 11 before falling below primitives alone at 20. Compression on P_{prim} removed more operations from primitive-only solutions over this range while held-out performance decreased, so solution compression alone was an incomplete measure of bounded-search value. Each added item made more low-operation programs available, which the solver examined before later programs where the item could help, and for the fixed one- and two-abstraction libraries, search budget determined whether the solver reached those later programs: increasing only the budget reversed the one-to-two loss, supporting the search-cutoff explanation for those libraries.

An abstraction's value must also cover how the abstraction was found. Search cost minimization's added selection work often required thousands of future problems to recover and did not recover in some cells. One factor stayed fixed throughout: the search process. Every result here comes from one deterministic solver, and a different search process could change the value of every library we measured. A practical system should therefore evaluate its complete library with its intended solver, budget, workload, and expected reuse. A useful abstraction must save more search than its selection and added programs cost on that workload.

Code and Data Availability

Code, generated data, and reproduction instructions are available in a public repository.

github.com/cucupac/program-synthesis.

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